Quantum Mechanics I — Exercise Sheet 3

Mixtures and the Bloch ball

Variant: Questions only

Conventions, as in Lecture 3: an ensemble $\{(p_k, \psi_k)\}$ has $p_k > 0$, $\sum_k p_k = 1$ and unit vectors ψ_k; its density matrix is $\rho = \sum_k p_k |\psi_k\rangle\langle\psi_k|$; for a qubit $\rho = \frac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma})$ with $r_a = \operatorname{Tr}\rho\sigma_a$; a branch with operator K has probability $\operatorname{Tr}(K\rho K^\dagger)$ and, if this is positive, leaves $K\rho K^\dagger/\operatorname{Tr}(K\rho K^\dagger)$.

Exercise 1 (all the two-member ensembles). Let $\mathbf{r} = (0, 0, 0.6)$. Show that for every point $\mathbf{r}_1$ of the Bloch sphere there is exactly one two-member ensemble of $\mathbf{r}$ that contains the state at $\mathbf{r}_1$: its other member is the second point $\mathbf{r}_2$ where the line from $\mathbf{r}_1$ through $\mathbf{r}$ meets the sphere, and the weight of $\mathbf{r}_1$ is $p = |\mathbf{r}_2 - \mathbf{r}|\,/|\mathbf{r}_2 - \mathbf{r}_1|$. Find the ensemble for $\mathbf{r}_1 = \mathbf{e}_z$ and for $\mathbf{r}_1 = (0.8, 0, 0.6)$.

Exercise 2 (three members). An ensemble has the states $|\uparrow\rangle$, $|{+}x\rangle$ and $|{+}y\rangle$, each with probability $\frac{1}{3}$. (i) Find ρ, its Bloch vector, its eigenvalues and $\operatorname{Tr}\rho^2$. (ii) A measurement along z reads +1. With what probability? Given this reading, with what probability was the system prepared in each of the three states?

Exercise 3 (how mixed can a state be?). Let ρ be the density matrix of an ensemble in $\mathbb{C}^n$. Show that $\operatorname{Tr}\rho^2 \geq 1/n$, with equality only for $\rho = \frac{1}{n}\mathbb{1}$. What does this say for a qubit?

Exercise 4 (readings nobody looks at). In this exercise a measurement is made but its reading is not recorded: every system is kept, whatever it read. By (3.8) and § 3.4, a qubit's Bloch vector $\mathbf{r}$ then becomes $(\mathbf{m}\cdot\mathbf{r})\,\mathbf{m}$. (i) Show that an unread reading along $\mathbf{m}$ never increases $|\mathbf{r}|$ or $\operatorname{Tr}\rho^2$. For which $\mathbf{r}$ does it leave ρ unchanged? (ii) An unread reading along z is followed at once by an unread reading along x. Show that every ρ ends at $\frac{1}{2}\mathbb{1}$. What is the state afterwards of a system whose second reading is known? (iii) ⋆ A system prepared in $|\uparrow\rangle$ undergoes, in turn, unread readings along $\mathbf{m}_k = (\sin(k\pi/N), 0, \cos(k\pi/N))$, $k = 1, \ldots, N$, with nothing acting in between. Find the probability that a reading along z at the end gives -1, its values for $N = 1$, 2 and 10, and its limit as $N \to \infty$.

Exercise 5 (tomography of a mixture). (i) Many systems are prepared alike, and the measured fractions of +1 along x, y, z are 0.7, 0.5, 0.5. Take them as the probabilities, and use Proposition 3.7 to find $\mathbf{r}$, ρ and $\operatorname{Tr}\rho^2$. Is the preparation pure? (ii) In another experiment the fractions along x, y, z are 0.9, 0.9, 0.9. Show that no density matrix gives these probabilities. What does this say about the fractions?

Exercise 6 ⋆. A qubit in any state ρ is measured along z. If the reading is -1, a resonant π-pulse of Lecture 2 is applied at once, which acts as $-i\sigma_x$; if it is +1, nothing is done. Write the two branch operators, and show that the density matrix afterwards, with the reading not recorded, is $|\uparrow\rangle\langle\uparrow|$ for every ρ.